

I. Introduction

As extreme drought events become more common, we can expect that plants will experience great water stress and as a result, phloem feeding herbivores like aphids will have fitness declines. Comparatively, it is uncertain how plant water stress from drought will indirectly affect the predatory insects feeding on the phloem feeding herbivores. Any fitness declines in aphids could potentially reduce their overall quality as food for higher trophic levels, especially parasitoid wasps that spend their entire juvenile development inside a living host. As a result, the goal of my research is to understand how changes in water stress (and subsequently aphid host quality) indirectly affect the overall success/fitness of the parasitoid wasp *Aphidius colemani*. I specifically want to understand how percent parasitism, or the number of mummies that form in a set number of aphids, varies with different degrees of water stress.

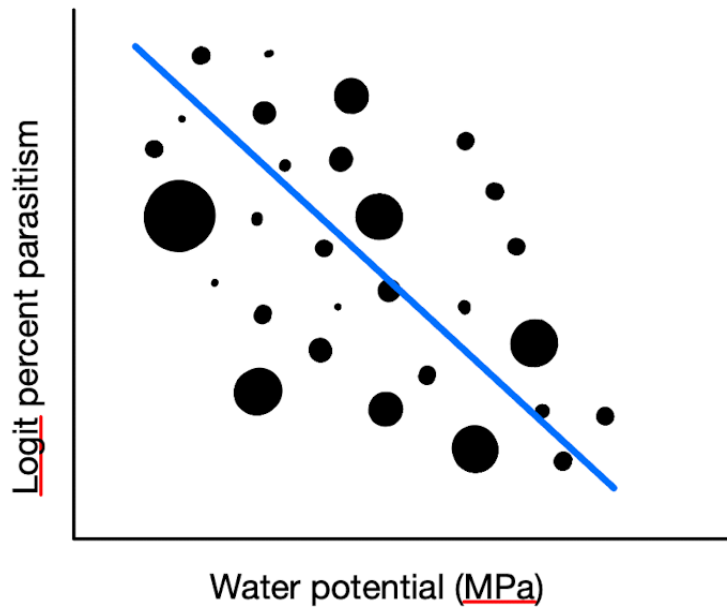
2. Methods

2.1. Study Design

Each experimental unit is a single BugDorm (mesh insect cage; Figure 1) with a water stressed plant in it that has been inoculated with 25 aphids 4 days before the experiment begins. 2 actively foraging female wasps were added to the cages and removed after 24 hours. The most important data collected is the Day 1 count of aphids and the total number of wasp pupae at the end.

The design has 28 replicates, or BugDorms. The treatment condition is the level of water stress, with each plant being dried down to a set percentage of field capacity weight between 100%-40%. Percentage difference was set based on the number of plants available and I made sure I had 2 of each percentage in case I had to drop one, so 100%-40% divided by 14. Treatments are intended not for direct comparison but instead the goal is to create a continuous gradient of water stress using the physiological measurement of plant water potential. There are 2 sets of 14 percentages ranging from 100%-40% as seen in the diagram. These were randomly distributed in the greenhouse as space allowed.

Example empirical logit plot:

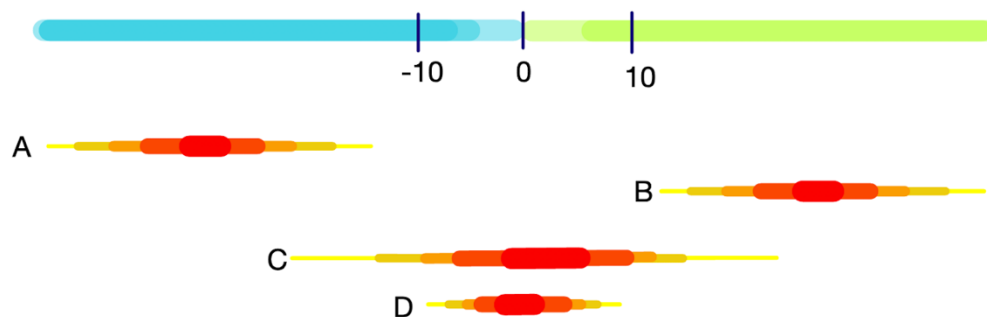


Steps for model iteration/refinement:

- Assess linearity and influential points using binned residual plots
- Run GLM with binomial distribution to assess relationship between percent parasitism and water potential
- Check for overdispersion. If over dispersed, use quasibinomial distribution.
- Use anova with χ^2 test to compare full and reduced model

Given the simplicity of the question, it is unlikely that this model will need to be refined in any way.

Quantitative Backdrop



The backdrop shows four intervals that correspond to four potential outcomes for the relationship between percent parasitism and plant water stress.

A. If the intervals (and resulting beta values) are on the negative end of the quantitative backdrop, it indicates that parasitism is increasing as plant water potential is more negative and plants are more water stressed. This would indicate that aphids feeding on water stressed plants are better hosts for developing parasitoids than aphids feeding on well-watered plants.

B. If the intervals (and resulting beta values) are on the positive negative end of the quantitative backdrop, it indicates that parasitism is decreasing as plant water potential is more negative and plants are more water stressed. This would indicate that aphids feeding on water stressed plants are worse hosts for developing parasitoids than aphids feeding on well-watered plants.

C. If the intervals (and resulting beta values) broadly cover both positive and negative variables, this means we are seeing both positive and negative trends in parasitism as plants are more water stressed. This could indicate that some other factor we did not control for is driving parasitism trends. It also could indicate that parasitism is affected by water stress nonlinearly, and that aphid host quality varies nonlinearly with plant water stress.

D. If the interval is around 0, this would indicate that parasitism is not very affected by plant water stress and is relatively the same regardless of water treatment. This would indicate that aphid quality as a host for the parasitoid wasps is consistent across varying degrees of plant water availability.

III. Hypothetical Results/Summary of Statistical Findings

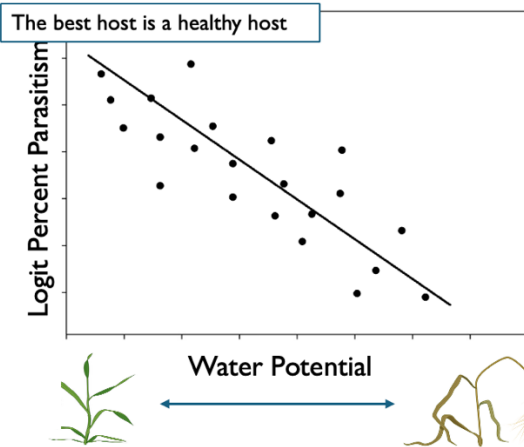
For model refinement, the most important step will be testing whether the data is overdispersed. If the dispersion parameter is greater than 1, a quasibinomial distribution will be used instead of a binomial distribution.

Hypothetical summary of statistics: Interval B, β_1 :

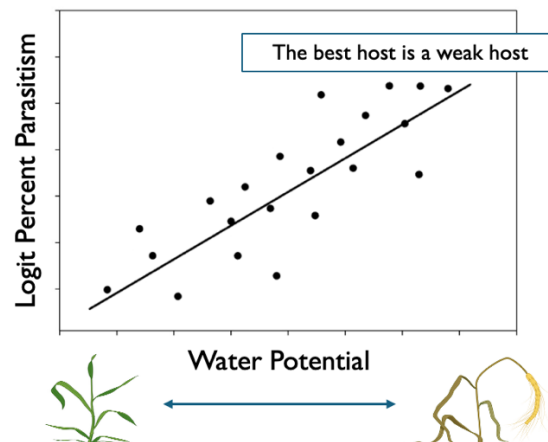
For every 1 MPa decrease in water potential, we see a 16 percent increase in aphid parasitism, and we are 95% confident that this ranges between a 13 to 19 percent increase in aphid parasitism by the parasitoid wasp *Aphidius colemani*. Because water potential is always a negative measurement and becomes more negative as plants have less water available, these data tell us that parasitism rates are declining as plants are more water stressed.

For assessing this result, I can make a ggplot scatterplot with a glm quasibinomial line of best fit. The examples below are just simple sketched scatter plots with a line of best fit, but it would look similar. A negative trend would indicate that aphids from well-watered plants are the best parasitoid hosts, and a positive trend would indicate that aphids from high-stress plants are the best parasitoid hosts.

Negative trend



Positive trend



IV. Scope of Inference

The results broadly apply to *A. colemani* wasps reared in captivity in *R. padi* aphids on wheat plants. Water stress level was randomly assigned, so we can assume that water stress as measured by water potential is causing any differences in parasitism that we observe. Given that the organisms were randomly sampled from the colony populations, we can assume that this applies broadly to this colony in a greenhouse or other controlled setting. I will likely use this work to discuss broader inferences about abiotic stress and trophic interactions broadly as this system is often used as a model system for similar questions.

V. Conclusion

If there is a negative association between plant water potential (a negative value) and percent parasitism, then parasitoid development is more successful in aphid hosts that are feeding on water stressed plants. Comparatively, if there is a positive association between plant water potential (a negative value) and percent parasitism, then parasitoid development is less successful in aphid hosts that are feeding on water stressed plants. If ultimately there seems to be consistent parasitism rates across a range of plant water potential, then parasitoid development numbers are unaffected by differences in plant water availability.

However, if there is a difference in percent parasitism, the next steps would be to assess both aphid composition and wheat composition changes across the same plant water potential gradient used for this experiment. We would look at lipid:protein ratios in aphids given the value of lipids and proteins for higher trophic levels like *A. colemani*. We would look at sugars and amino acids in the wheat plants because that is what the aphids are feeding on. These experiments will help us better explain why we are seeing a change in parasitism rates with differences in water stress and if these changes are host mediated.